

## Week 4

```
For more details see ps(1).
rishi_alapati@BXTHub:~$
rishi_alapati@BXTHub:~$ nano q4.c
rishi_alapati@BXTHub:~$ gcc q4.c -o q4
./q4
Child 1: PID = 14480
Child 2: PID = 14481
Child 3: PID = 14482
Child 1 finished
Child 2 finished
Child 3 finished
All children completed
rishi_alapati@BXTHub:~$ nano zombie.c
rishi_alapati@BXTHub:~$ gcc zombie.c -o zombie
rishi_alapati@BXTHub:~$ ./zombie
Parent PID: 14528
Child PID: 14529
Child finished
ps -el | grep zombie
wait(NULL);
Zombie removed
rishi_alapati@BXTHub:~$ ps -el | grep zombie
rishi_alapati@BXTHub:~$ wait(NULL);
-bash: syntax error near unexpected token 'NULL'
rishi_alapati@BXTHub:~$ gcc q1.c -o q1
gcc q2.c -o q2
gcc q3.c -o q3
gcc q4.c -o q4
gcc zombie.c -o zombie
rishi_alapati@BXTHub:~$ ./q1
./q2
./q3
./q4
./zombie
Enter Linux command: ls
Parent PID: 14640
Child PID: 14641
a.out copy.txt p1.c q1 q1.c q2 q2.c q3 q3.c q4 q4.c rishi.c source.txt zombie zombie.c
Child completed
File copied successfully
Parent PID: 14653
Child PID: 14654
```

```
gcc zombie.c -o zombie
rishi_alapati@BXTHub:~$ ./q1
./q2
./q3
./q4
./zombie
Enter Linux command: ls
Parent PID: 14640
Child PID: 14641
a.out copy.txt p1.c q1 q1.c q2 q2.c q3 q3.c q4 q4.c rishi.c source.txt zombie zombie.c
Child completed
File copied successfully
Parent PID: 14653
Child PID: 14654
Child PID: 14654
Parent PID: 14653
Child finished
Parent finished
Child 1: PID = 14667
Child 2: PID = 14668
Child 3: PID = 14669
Child 1 finished
Child 2 finished
Child 3 finished
All children completed
Parent PID: 14677
Child PID: 14678
Child finished
Zombie removed
rishi_alapati@BXTHub:~$
```

```
pranav_chowdary@Pranav: ~  
GNU nano 8.7.1 q4.c  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    int i;  
  
    for (i = 0; i < 3; i++)  
    {  
        if (fork() == 0)  
        {  
            printf("Child %d: PID = %d\n", i + 1, getpid());  
            sleep(i + 1);  
            printf("Child %d finished\n", i + 1);  
            return 0;  
        }  
    }  
  
    for (i = 0; i < 3; i++)  
    {  
        wait(NULL);  
    }  
  
    printf("All children completed\n");  
  
    return 0;  
}
```

[ Read 28 lines ]

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|----------------|---------------------|--------------------|-----------------|-------------------|----------------------|-----------------|---------------------|
| <b>^G</b> Help | <b>^O</b> Write Out | <b>^F</b> Where Is | <b>^K</b> Cut   | <b>^T</b> Execute | <b>^C</b> Location   | <b>M-U</b> Undo | <b>M-A</b> Set Mark |
| <b>^X</b> Exit | <b>^R</b> Read File | <b>^N</b> Replace  | <b>^U</b> Paste | <b>^J</b> Justify | <b>^_</b> Go To Line | <b>M-E</b> Redo | <b>M-G</b> Copy     |